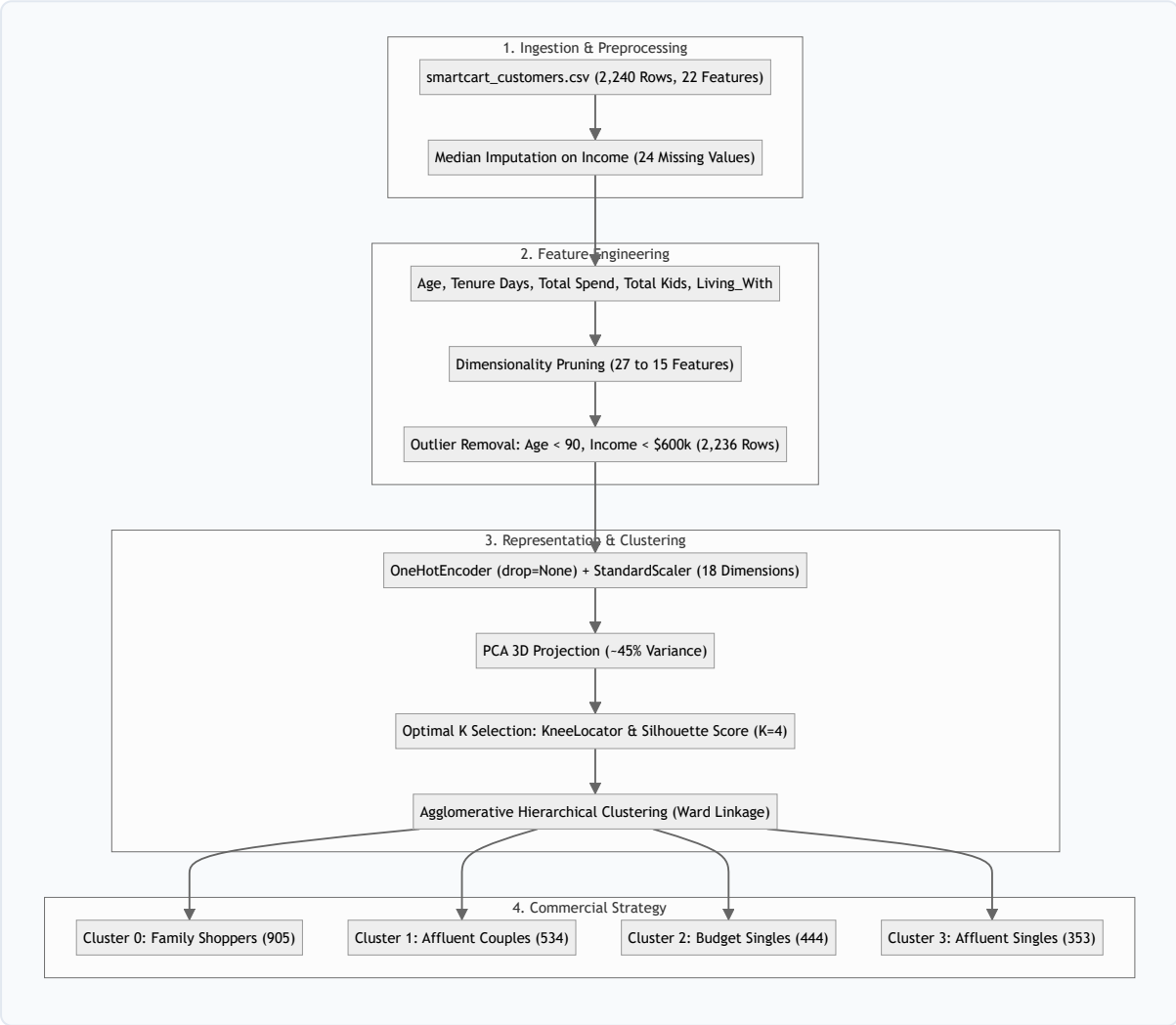
Lecture 33: SmartCart Clustering System (End-to-End Minor Project)

Course Module Overview

Welcome to **Lecture 33**, a comprehensive minor project applying **Unsupervised Machine Learning** to enterprise e-commerce customer segmentation. Throughout this project, we transition from raw transaction data to an operational clustering pipeline delivering four data-backed customer personas and targeted commercial marketing playbooks.



Submodule Architecture & Directory Index

Submodule	Topic & Technical Domain	Official Code & Assets Reference
Notes 33.01	Project Introduction, Business Objectives & Exploratory Audit	Cells 00 - 04 in `smartcart.ipynb`
Notes 33.02	Handling Missing Values: Statistical Median Imputation on Income	Cells 05 - 08 in `smartcart.ipynb`
Notes 33.03	Feature Engineering: Domain Synthesis of Age, Tenure, Spend & Family	Cells 09 - 15 in `smartcart.ipynb`
Notes 33.04	Dropping Redundant Columns & Dimensionality Management	Cells 16 - 20 in `smartcart.ipynb`
Notes 33.05	Handling Outliers: Pairplots & Demographic/Financial Bounds	Cells 21 - 23 in `smartcart.ipynb`
Notes 33.06	Feature Correlation Matrix & Pearson Heatmap Analysis	Cells 24 - 28 in `smartcart.ipynb`
Notes 33.07	Categorical Feature Encoding (OneHotEncoder) & Scaling (StandardScaler)	Cells 29 - 39 in `smartcart.ipynb`
Notes 33.08	High-Dimensional Data Visualization: PCA 2D vs 3D Projections	Cells 40 - 45 in `smartcart.ipynb`
Notes 33.09	Optimal Cluster Selection: KneeLocator (Elbow Method) & Silhouette Score	Cells 46 - 53 in `smartcart.ipynb`
Notes 33.10	Clustering Execution: K-Means vs Agglomerative Hierarchical Clustering	Cells 54 - 59 in `smartcart.ipynb`
Notes 33.11	Cluster Characterization: Income vs Spending Behavioral Patterns	Cells 60 - 64 in `smartcart.ipynb`
Notes 33.12	Cluster Summary Matrix & Four Customer Persona Profiling Playbooks	Cells 65 - 66 in `smartcart.ipynb`

Submodule	Topic & Technical Domain	Official Code & Assets Reference
SmartCart_Project	Complete Project Hub: Jupyter Notebook, Raw Dataset & Assignment Brief	`smartcart.ipynb` , `smartcart_customers.csv`
Practice_Problems	Three Industry Case Studies with Complete Mathematical Solutions	`Solutions.md`

Production Persona Summary ($N = 2,236$ Customers)

- **Cluster 0 (40.5% - 905 Customers)**: *"Family Shoppers / Thrifty Couples"* — Moderate income (\$39.7k), moderate spend (\$222), multiple children (1.24), discount coupon seekers.
- **Cluster 1 (23.9% - 534 Customers)**: *"Affluent Couples / Established Spenders"* — High income (\$72.8k), high spend (\$1,237), low children (0.51), dominant physical store and catalog buyers.
- **Cluster 2 (19.9% - 444 Customers)**: *"Budget Singles / Window Shoppers"* — Lowest income (\$37.0k), lowest spend (\$166), single parents (1.27 kids), highest web browsing with low conversion.
- **Cluster 3 (15.8% - 353 Customers)**: *"Affluent Singles / Golden Cluster"* — High income (\$70.7k), high spend (\$1,190), living alone, **Highest Campaign Response (32.0%)** delivering maximum marketing ROI.